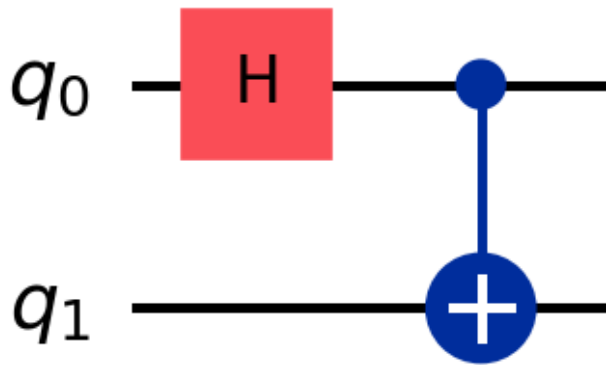


FIRST LAB EXERCISE

Quantum gates, Bell states and the no-cloning theorem



Python implementation with Qiskit

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1. Objective and software environment

This practice reproduces the experiments proposed for QUIRK by using Python and Qiskit. The program constructs quantum circuits, obtains exact state vectors, computes reduced density matrices and produces figures that can be compared with the Bloch-sphere and probability views in QUIRK.

Component	Use
Python 3	Main programming language
Qiskit	Quantum circuits and state-vector calculations
NumPy	Matrix verification and numerical operations
Matplotlib	Plots and circuit figures

2. One-qubit gates and state preparation

The products $X.X$ and $H.H$ are the identity. The square-root of X gate used in the script also satisfies $\text{sqrt}(X).\text{sqrt}(X)=X$. The residual norms are at floating-point precision.

Verification	Numerical error
$ X.X-I $	0.000e+00
$ H.H-I $	3.156e-16
$ \text{sqrt}(X)^2-X $	0.000e+00

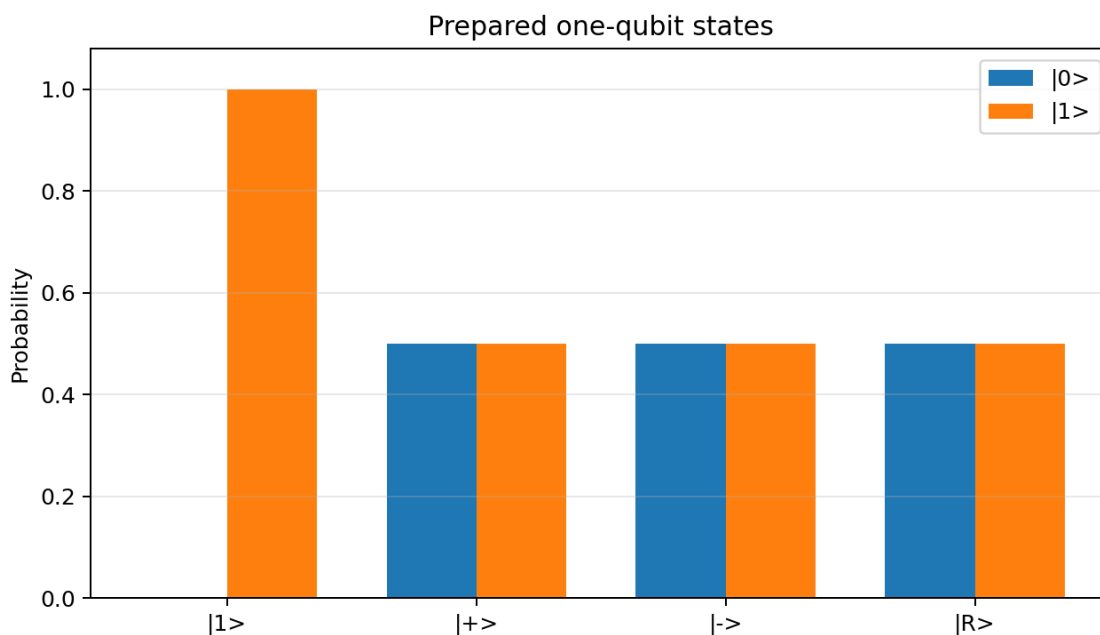


Figure 1. Computational-basis probabilities of $|1\rangle$, $|+\rangle$, $|-\rangle$ and $|R\rangle$.

The states $|+\rangle$, $|-\rangle$ and $|R\rangle$ have equal probabilities in the computational basis. Their differences are relative phases, which are visible in their Bloch vectors but not in a direct Z-basis measurement.

2.1 Python implementation

```
qc = QuantumCircuit(1, name="|->")
qc.h(0)
qc.z(0)
circuits["|->"] = qc

qc = QuantumCircuit(1, name="|R>")
qc.h(0)
```

```

qc.s(0)
circuits["|R>"] = qc
return circuits

def bell_circuits() -> dict[str, QuantumCircuit]:
    circuits: dict[str, QuantumCircuit] = {}
    for name in ("Phi+", "Phi-", "Psi+", "Psi-"):
        qc = QuantumCircuit(2, name=name)
        qc.h(0)
        qc.cx(0, 1)
        if name in ("Phi-", "Psi-"):
            qc.z(0)
        if name in ("Psi+", "Psi-"):
            qc.x(1)
        circuits[name] = qc
    return circuits

def controlled_truth_table(gate_name: str, anti_control: bool = False) -> dict[str, str]:
    table: dict[str, str] = {}
    gate = XGate() if gate_name == "X" else HGate()
    controlled = gate.control(1, ctrl_state=0 if anti_control else 1)
    for value in range(4):
        qc = QuantumCircuit(2)
        if value & 1:
            qc.x(0)
        if value & 2:
            qc.x(1)
        qc.append(controlled, [0, 1])

```

3. Controlled and anti-controlled operations

The controlled-X gate changes the target only when the control is $|1\rangle$. The anti-controlled version changes it when the control is $|0\rangle$. Replacing X by H follows the same control rule, although an H operation can create a superposition rather than simply flipping a bit.

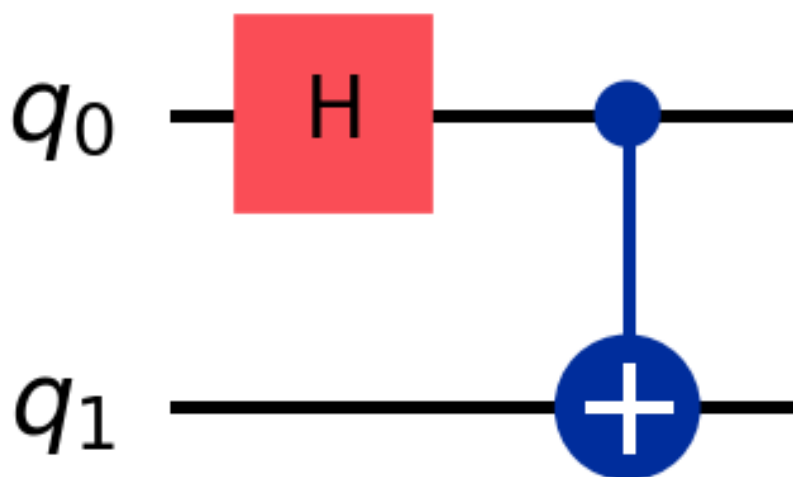


Figure 2. Hadamard followed by CNOT, the standard Bell-state preparation block.

4. Bell states

All four Bell states were generated from $|00\rangle$. A Hadamard gate creates a balanced superposition on the first logical qubit and the CNOT correlates both qubits. Additional X and Z gates select the required Bell state.

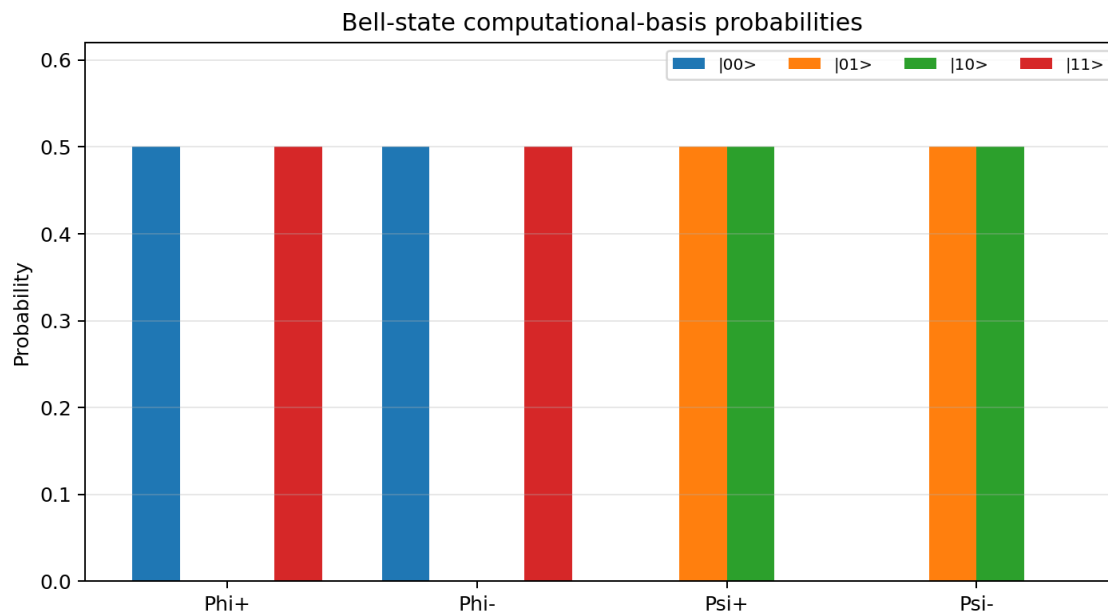


Figure 3. Ideal Bell-state probabilities. Relative signs do not alter computational-basis counts.

5. No-cloning theorem

Applying CNOT to $|+\rangle|0\rangle$ produces $(|00\rangle + |11\rangle)/\sqrt{2}$, not $|+\rangle|+\rangle$. The reduced state of either qubit has purity 0.5, which identifies a maximally mixed local state. The information is stored in correlations between the qubits, so the output is entangled rather than two independent copies.

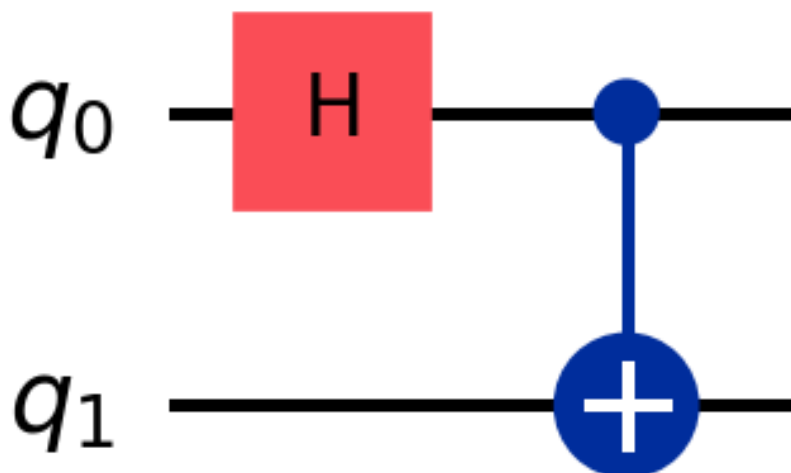


Figure 4. CNOT applied to $|+\rangle|0\rangle$.

Reduced state	Purity
Control	0.500
Target	0.500

6. Individual qubits and entanglement

Register A is the separable state $|+\rangle|+\rangle$, whereas register B is the Bell state Φ^+ . Before an additional Hadamard, every qubit has $P(1)=0.5$. Nevertheless, their reduced density matrices are different: the qubits of register A are pure, while each qubit of register B is maximally mixed. This explains why identical Bloch-sphere probabilities do not imply identical two-qubit states.

Qubit	P(1)	Purity	Bloch vector
A1	0.500	1.000	1.00, 0.00, 0.00
A2	0.500	1.000	1.00, 0.00, 0.00
B1	0.500	0.500	0.00, 0.00, 0.00
B2	0.500	0.500	0.00, 0.00, 0.00

7. Conclusions

The Python implementation confirms the gate identities, prepares the requested one-qubit states, constructs all Bell states and demonstrates that CNOT cannot clone an arbitrary quantum state. Local measurements alone are insufficient to describe an entangled system; the joint state and its correlations are essential.

8. Files and execution

Run `python iqcl_quantum_fundamentals.py`. The script writes `IQC1_results.json` and regenerates all images in the `figures` folder. The notebook provides an alternative interactive entry point.